

Eine Woche, ein Beispiel

4.26. moduli of v.b. over curve

This document is mainly a table of explicit description for moduli of v.b. over curve.

Ref:

[Beau94]: Arnaud Beauville. "Vector Bundles on Curves and Generalized Theta Functions: Recent Results and Open Problems." arXiv:alg-Geom/9404001. Preprint, arXiv, April 5, 1994.
<https://doi.org/10.48550/arXiv.alg-geom/9404001>.

[Beau05]: Arnaud Beauville. "Vector Bundles on Curves and Theta Functions." arXiv:math/0502179. Preprint, arXiv, February 11, 2005. <https://doi.org/10.48550/arXiv.math/0502179>.

For a sm proj curve $C/\mathbb{C}$ with $g(C)=g$,
we want to describe

$$\mathcal{M}_C(r, \mathcal{O}_C) = \{ \text{semistable v.bs on } C \text{ with rk } r \} / \sim$$

Also written as $\text{SU}_C(r)$

One can construct a rational map

$$\begin{aligned} \varphi: \mathcal{M}_C(r, \mathcal{O}_C) &\dashrightarrow |\mathcal{H}_C| \cong \mathbb{P}^{r^2-1} \\ \varepsilon &\longmapsto \mathcal{H}_\varepsilon = \{ \mathcal{L} \in \text{Pic}^{g-1}(C) \mid H^0(\varepsilon \otimes \mathcal{L}) \neq 0 \} \end{aligned}$$

Well-definedness:

- ① Writing $\mathcal{H}_\varepsilon$ as the determinant loci,
we get $\mathcal{H}_\varepsilon$ is of $\text{codim} \leq 1$;
- ② When $\mathcal{H}_\varepsilon$ is a divisor,
use Porteous' formula to show that $[\mathcal{H}_\varepsilon] = [r\mathcal{H}_C]$;
- ③ verify that $\varphi(\mathcal{O}_C^{\oplus r}) = r\mathcal{H}_C$ is a divisor.

Conj. [Beau 05, Prop 4.5]
 φ is generically finite.

$g \backslash r$	1	2	3	4
1	0 0	1 1 1 $\mathbb{P}^1 \subset \text{Sym}^2 E$	1 2 2 $\mathbb{P}^2 \subset \text{Sym}^3 E$	1 3 3 $\mathbb{P}^3 \subset \text{Sym}^4 E$ 1
2	0 0	1 3 3 $\mathbb{P}^3$ 1	2 8 8 ramified over deg 6 hypersurface 2	15 15 gen finite not finite
3	0 0	2/1 6 7 Coble quartic 4	2/1 16 26 finite ≤ 54	2/1 30 63
4	0 0	2/1 9 15 96	2/1 24 80 ≤ 810	2/1 45 255
...				

$\Leftarrow$
 proved
 to be regular map

$2/1$:
 $\deg \varphi = \begin{cases} 2, & \text{hyperelliptic} \\ 1, & \text{otherwise} \end{cases}$

$\deg \varphi$	$(g-1) \binom{r^2-1}{2}$ $\dim \mathcal{M}_g(r, \mathcal{O}_C)$	$r^g - 1$ $\dim r\Theta $
explicit description in terms of φ	$\deg \varphi \cdot \deg \text{Im } \varphi$	